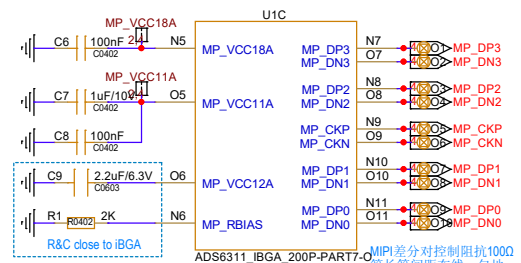
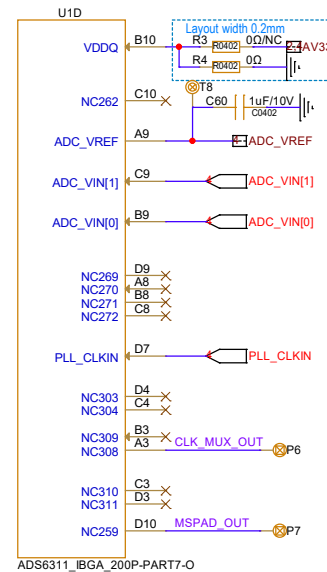
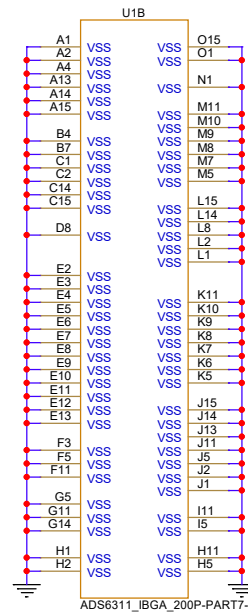
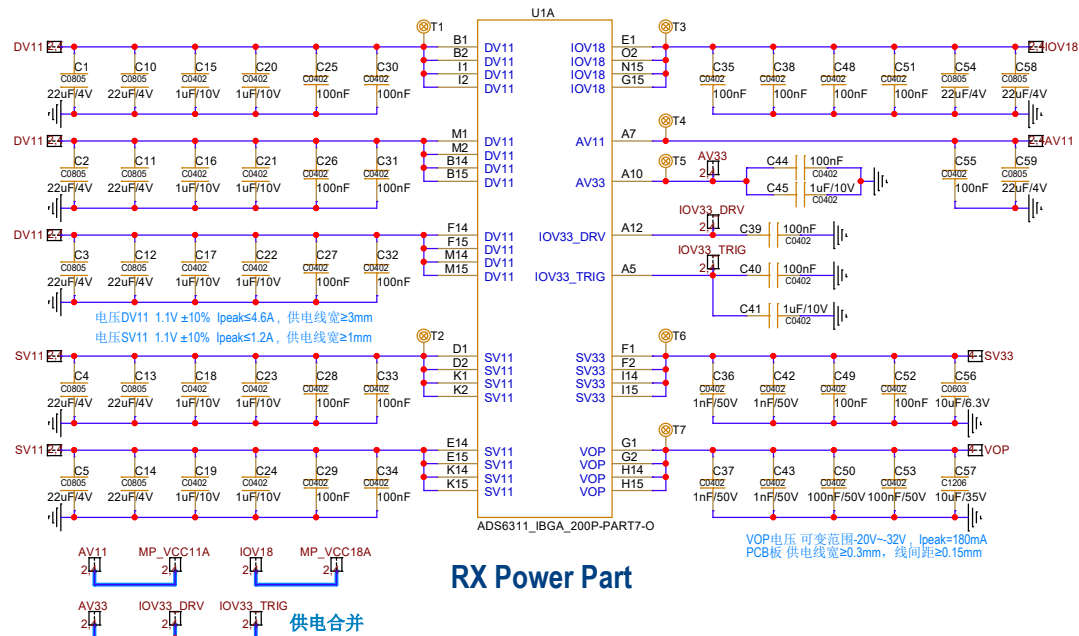


原理图设计变更

版本	修改人	日期	修改项	修改内容
1V0	HY	202400823	1. RX板与TX板合并	a. RX板与TX板合并为一个PCB板，最小化面积 b. 采用iBGA封装sensor芯片和1D 32ch VCSEL
			2. RX部分修改	a. 电压VOP和SV33将两个1uF电容改为1个10uF b. VDDQ引进默认接到GND，预留接AV33烧eFuse c. 模组时钟默认采用板上有源晶振提供 d. 引出功能安全时钟 TREFCLK信号
			3. TX部分修改	a. 两个选通芯片与NOR FLASH合并一组SPI通信 b. 比较DAC调整为电阻分压提供 c. 温度传感器芯片调整为NTC测VCSEL温度

硬件框架图

Project Title	Hawk_LIDAR_RX-TX_Minimodule		 ADAPS PHOTOGRAPHY
Page Name	1_Block_Diagram		
Designed by	H_Yang	NO.	HAWK_Minimodule
Checked by	LJW	Rev.	1V0
Approved by	JFB	Size	A3
Modify Date	Friday, October 11, 2024	Sheet	1 of 4



推荐电容型号

电容: 22uF/4V/X7T/0805
GCM21BD70G226ME36L

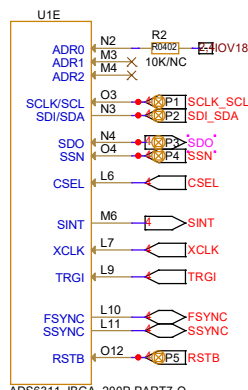
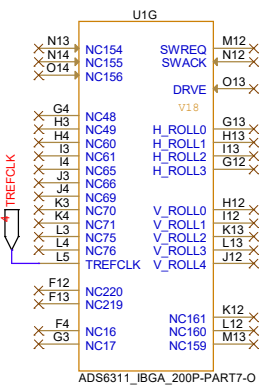
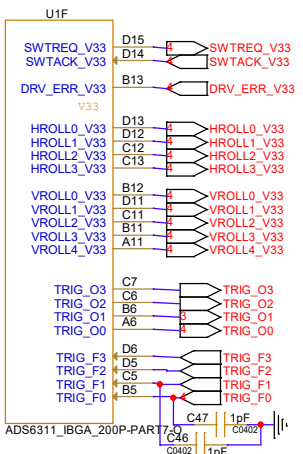
电容: 10uF/35V/X7R/1206
GCM31CD7YA106KE36L

电容: 100nF/50V/X7R/0402
GCM155R71H104KE02D

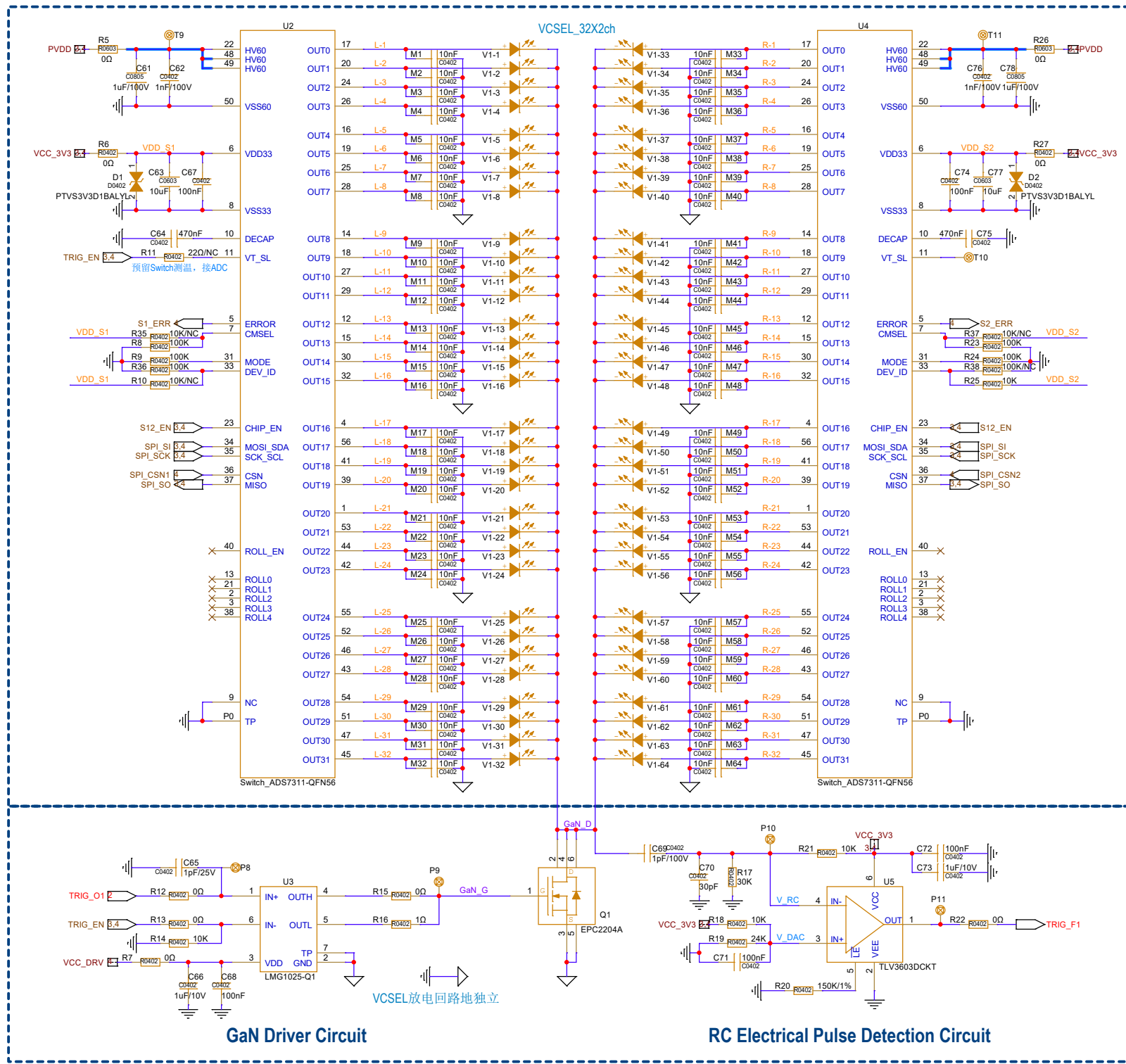
电容: 2.2uF/6.3V/X7R/0603
GCM188R70J225KE22D

电容: 1uF/10V/X7S/0402
GCM155C71A105KE38D

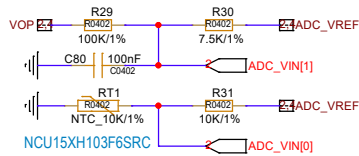
电容: 1nF/50V/X7R/0402
GCM155R71H102KA37D



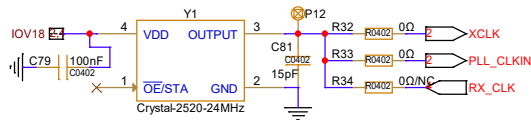
Project Title	Hawk_LIDAR_RX-TX_MinModule			
Page Name	2_RX_IBGA_200P_Chip			
Designed by	H_Yang	NO.	HAWK_MinModule	
Checked by	LJW	Rev.	1V0	
Approved by	JFB	Size	A3	
Modify Date	Friday, October 11, 2024	Sheet	2 of 4	



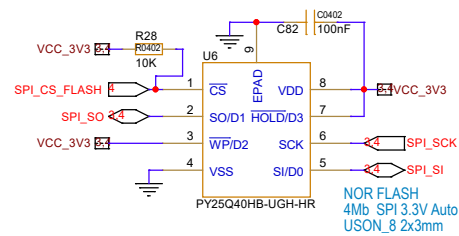
Project Title	Hawk_LIDAR_RX-TX_Minimodule			
Page Name	3_TX_Driver_Circuit			
Designed by	H_Yang	NO.	HAWK_Minimodule	
Checked by	LJW	Rev.	1V0	
Approved by	JFB	Size	A3	
Modify Date	Friday, October 11, 2024	Sheet	3 of 4	



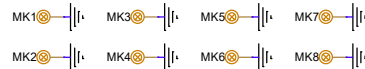
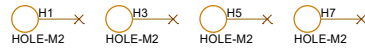
ADC Detection



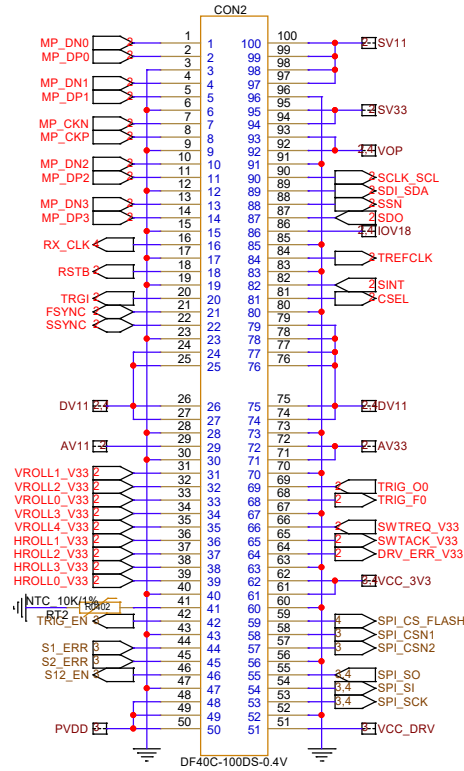
Crystal Oscillator Supply



NOR FLASH



PCB Fixed hole



对应母座：DF40HC(3.0)-100DS-0.4V

RX_TX MiniModule Connect

Project Title	Hawk_LIDAR_RX-TX_Minimodule		
Page Name	4_Module_Connect		
Designed by	H_Yang	NO.	HAWK_Minimodule
Checked by	LJW	Rev.	1V0
Approved by	JFB	Size	A3
Modify Date	Monday, October 21, 2024	Sheet	4 of 4